Ch. 30 CW

Principles of Macroeconomics

March 3, 2026

1. Suppose that the reserve requirement for checking deposits is 10 percent and that banks do not hold any excess reserves.

   (a) If the Fed sells $1 million of government bonds, what is the effect on the economy's reserves and money supply?

   (b) Now suppose that the Fed lowers the reserve requirement to 5 percent but that banks choose to hold another 5 percent of deposits as excess reserves. Why might banks do so? What is the overall change in the money multiplier and the money supply as a result of these actions?

2. Assume that the banking system has total reserves of $100 billion. Assume also that required reserves are 10 percent of checking deposits, banks hold no excess reserves, and households hold no currency.

   (a) What is the money multiplier? What is the money supply?

   (b) If the Fed now raises required reserves to 20 percent of deposits, what are the change in reserves and the change in the money supply?

3. Assume that the reserve requirement is 20 percent. Also assume that banks do not hold excess reserves and that the public does not hold any cash. The Fed decides that it wants to expand the money supply by $40 million.

   (a) If the Fed is using open-market operations, will it buy or sell bonds?

   (b) What quantity of bonds does the Fed need to buy or sell to accomplish the goal? Explain your reasoning.

4. The economy of Elmendyn contains two thousand $1 bills.

   (a) If people hold all money as currency, what is the quantity of money?

   (b) If people hold all money as demand deposits and banks maintain 100 percent reserves, what is the quantity of money?

   (c) If people hold equal amounts of currency and demand deposits and banks maintain 100 percent reserves, what is the quantity of money?

   (d) If people hold all money as demand deposits and banks maintain a reserve ratio of 10 percent, what is the quantity of money?

   (e) If people hold equal amounts of currency and demand deposits and banks maintain a reserve ratio of 10 percent, what is the quantity of money?

5. (a) Bank reserves are 200; the public holds 200 in currency, and the reserve ratio is 0.25. Find deposits and the money supply.

   (b) The money supply is 650, and currency held by the public equals bank reserves. The reserve ratio is 0.25. Find currency held by the public and bank reserves.

   (c) The money supply is 1,500 of which 400 is currency held by the public. Bank reserves are 110. Find the reserve ratio.